

Article type

Development of an improvising machine in an ubimus context

Abstract

This paper explores the relationship between design decisions in improvising machines and the ideological foundations of ubimus movement practices. To this end, we present a *free/libre* open-source Pure Data interactive system, named “XXXXXXXXXX”, initially designed to evaluate the potential of learning free improvisation through interaction with improvising machines in the context of social distancing experienced by the post-2020 world. Our system is now envisioned as a possible creative tool with applicability in areas other than improvisation, such as composing electronic music, extending digital instruments, and its use in aesthetic proposals that are not necessarily related to free improvisation and will be discussed in further detail. Our system's free software and do-it-yourself approach inspire musicians to explore the creative potential of decentralized technological interaction. This is especially relevant in a context where intelligent agents, often poorly understood and controlled, are increasingly pervasive in our lives.

Keywords: ubimus; interactive music systems; improvising machines; free improvisation; computer creativity.

1. Introduction

The ubiquitous music movement (ubimus) arises from the convergence of musical creativity, ubiquitous computing tools, and the social dynamics of participant interaction, redefining how music is created and experienced^[1]. Ubimus has been developing from its initial characteristics by integrating new musical activities and perspectives that may even challenge the necessity of technological support. As a result, the definitions of ubimus have been updated regularly. Recent publications suggest that a consensus for a definition has been solidified in the “notion of ubimus as an ecology or ecosystem of stakeholders and resources that support creative endeavors”^[2].

While not particularly circumscribed as a ubimus practice, previous initiatives in the area have been related to the genre of free improvisation. Stolfi *et al.*^[3] describes the development of an

online music-making tool that takes advantage of the Freesound library¹ to make real-time selections and combinations of Creative Commons audio samples in the context of live improvisations. In another example, Clemente *et al.*^[4] conducted a series of experiments with Information and Communication Technologies (ICT) and Ubiquitous Computing (UC) as tools to support free improvisation practices of a saxophone ensemble, observing improvements in the group's sound quality, according to the students and teachers who took part in the experiment.

Free improvisation can be understood as a musical practice associated with an anti-formalist approach to music-making, where musical pieces are completely improvised at the moment, abstaining from the explicit use of conventional musical structures and techniques in favor of the exploration of sonic materials that arise from the interplay between musicians. The difficulty of studying free improvisation, evidenced by the obstacles found in suggestions for its formalization^[5], suggests that other paths are necessary for its understanding and, ultimately, apprehension. While free improvisation avoids predetermined structures, performers still require preparation for public performances. As we understand it, many improvisers suggest the practice of repeated ensemble performances to be the most fruitful methodology for musicians to delve into the free improvisation universe^[6-9].

This scenario begs the question: if ensemble practice proves to be the primary learning approach for free improvisation, how can musicians without access to companionship for improvisation sessions achieve some level of proficiency in this genre? In this sense, the pandemic world of post-2020 has offered a challenging scenario for musicians who want to practice in-person musical activities. However, social distancing also opened doors to the exploration of connected forms of musical creation for a broader pool of participants, not previously engaged in this sort of musical experience.

Technological solutions proposed by companies for the *musical internet* typically try to encompass a wide range of musical applications for a single product, often reinforcing the acoustic-instrumental model of musical production and a hierarchical approach to musical interaction. In the other direction, Messina *et al.*^[10] stress the ability of ubimus ecosystems to adapt their actions to local conditions, leveraging the flexibility of available resources and

¹ Available at <https://freesound.org/>.

approaching the presence of technological resources not just as passive instruments but as active stakeholders in the creative process.

This paper discusses the intersection between free improvisation and ubimus practices as a fertile ground for exploring interactive systems that demonstrate a sense of independence and agency when improvising with musicians. Tatar and Pasquier^[11] define *musical agents* as systems that “explore the notions of autonomy, reactivity, proactivity, adaptability, coordination and emergence” in interaction with an environment of musical creativity, ultimately reducing their creative tasks into twelve categories. This paper aims to draw connections between the reasoning behind the development of improvising machines, one of those categories, and the ideological principles underpinning ubimus practices. To do so, we discuss the latest decisions in the design, implementation, and future perspectives for our improvising machine, named [REDACTED], a *free/libre* open-source software² (FLOSS) project developed in Pure Data (Pd).

This work seeks to contribute to the broader discussion on the role of artificial agents in musical discourses, particularly in the context of live, interactive performance, and the unique challenges posed by integrating machine intelligence into human-centered musical ecosystems. We believe that the do-it-yourself (DIY) production process and the open distribution nature of our system are capable of helping other musicians in the endeavor to create new creative solutions both in the field of free improvisation and ubiquitous music.

2. The system

[REDACTED] is a musical agent originally designed for live performances of free improvisation and approached in previous work^[12] as a feasible companionship for practicing the genre. This section presents its later version, with novel solutions for each step of the interaction.

Like other musical agents for live performances, [REDACTED] analyzes the musician's input using music information retrieval (MIR) methods. The system then translates the analyzed data into meaningful musical responses. In our particular approach to the interaction process, [REDACTED] stores this data as “phrases” to be manipulated by mathematical operations for the construction

² Available at [https://github.com/\[REDACTED\]](https://github.com/[REDACTED]).

of new musical discourses. The sonic responses of the machine can be delivered as real-time synthesized audio or MIDI messages sent to any digital music instrument.

In the first step of the interaction model, [REDACTED] analyzes sounds played by the musician with whom it interacts, segmenting these sounds into “phrases”, identified as a list of sequential sonic events between two moments of silence. Each musical parameter extracted from a sonic event is added to separate lists. Therefore, a musical phrase is disassembled into several lists that represent a single musical parameter.

Using the k-means algorithm, an unsupervised clustering method, the system categorizes sound parameters into user-defined groups for analysis. The musical parameters timbre, register, intensity, and tempo are categorized by the k-means into varying numbers of clusters (in ten iterations), which we believe could potentially represent a simplified perception of these musical parameters by musicians. This model of sound analysis and clustering aims to enable the interaction to be focused on a process that is potentially natural to musicians, whose expectations of musical interaction usually revolve around their peers' ability to listen.

Each sound parameter has a numerical value to represent their cluster, corresponding to: (0) non-percussive and (1) percussive sound for timbre; (0) low, (1) middle, and (2) high registers; (0) piano, (1) mezzo and (2) forte intensities; and (0) long, (1) moderate and (2) short for sound duration. Lists comprising the cluster identified for each sound event are recorded in distinct datasets³, which constitute collections of all lists of parameters (corresponding to phrases) played by the musician during past and current improvisation sessions, essentially functioning as long-term memory. To match a phrase to a specific dataset, we use the statistical mode found for the phrase, following the same clusters for sound events⁴, approximating how a musician might perceive prevalent aspects of a musical texture. For example, in a sequence of many short notes and one long note, the prevailing characteristic will be understood as short durations.

As soon as the identified cluster for a musical parameter changes, meaning that the musician has altered some prevalent aspect of the musical texture being played, [REDACTED] randomly selects a list from a dataset corresponding to the same parameter. To decide which dataset to choose from, [REDACTED] makes use of *second order* Markov chains to model the likelihood of the cluster for

³ Simple .csv files.

⁴ Thus providing two datasets for timbre, and three for pitch, intensity and tempo each.

the following phrase. As suggested by Roads^[13], Markov chains calculate the probability of occurrence of the next value in the sequence considering only the state in which the system is located (zeroth order). For musical applications, considering the last state (first order), or even the last two states (second order), would enable the construction of sequences with a more organized structural sense. As the order increases, the material generated by the algorithm approaches the original content of the sequence. This method has been periodically used in musical agents that perform imitative tasks, statistically modeling the playing style of a particular human whom it interacts with, generally concerning rhythmic and melodic aspects^[14].

In turn, [REDACTED]'s use of Markov chains does not serve the purpose of “imitating” what is played by the user, but rather looking for trends in behavior between each phrase played. For example, once a musician repeats a pattern, such as playing notes in the high register, followed by notes in the low register, the machine will tend to replicate this behavior with greater probability. However, the phrase played by the machine may differ greatly from the material drawn from “memory”, since each musical parameter of the original phrase has been disassociated from the others, and transformative operations will modify the symbolic data found in the list selected from the dataset.

Initially programmed with five voices, [REDACTED] now only has a monophonic texture to respond to the user. The choice to simplify interaction to just one voice aims to direct the user's listening experience to more concise musical content, making it easier to understand what the machine is playing. Two processes for transforming the musical material drawn from memory can be used to generate new content of the musical response: *homorhythm* and *counterpoint*.

A homorhythmic texture⁵ is activated based on identifying a low standard deviation for the last few inter-onset intervals (IOI) played by the musician. Even though sophisticated methods have been proposed for beat identification, this simple yet effective method allows the system to quickly perceive whether the musician is approaching the literal reproduction of a pulse, promptly joining the beat. Although handled by a different algorithm, Collins^[15] has proposed, similarly, a rhythmic analysis to “differentiate free time playing from steadier metrically established performance”, inspiring the decision for the two modes of texture interaction

⁵ That is, when the machine makes sounds that start at the same time as the sounds made by the musician with whom it interacts.

implemented in [REDACTED]. Musical parameters for each note played during the homorhythmic texture are extracted not from memory but from the last phrase played, bringing the machine's sound material closer to that which has just been played by the human. The system exits the homorhythmic mode once the time elapsed since its activation exceeds the duration of the most recently played phrase.

When no steady pulse is identified, [REDACTED] plays a contrapuntal texture, in which lists of parameters drawn from memory are manipulated and combined to formulate new phrases to be played by the machine. As each sound parameter is dissociated from the others found in its original phrase, and manipulative operations are applied to the list, the resulting texture is unlikely to be heard as directly related to a phrase played by the musician. To manipulate each list of parameters, the contrapuntal method randomly selects from a range of mathematical functions to be applied to the numerical values of the list taken from memory. For rhythm, these functions and their explanations can be seen in Table 1. Manipulations of pitches represent notes by their pitch classes^[16](PC), and the functions to manipulate them are presented in Table 2. Finally, Table 3 presents the functions for amplitude manipulations.

Table 1. Functions that can be applied to rhythmic manipulation of recorded phrases.

Same	Bypasses the original list with no modification.
Accelerando	Rearranges the list of durations in descending order of values.
Ritardando	Rearranges the list of durations in ascending order of values.
Shuffle	Shuffles the order of the durations in the original list.
Multiple	Multiplies the duration values by a random factor.
Addition	<i>Added Value Rhythm</i> as described by Messiaen ^[17] .
Ostinato	Chooses a duration and repeats it for the number of items in the list.
Rotation	Rotates the list for a random number of iterations.
Reversion	Reverses the list's order.
Random	Creates a new list of random values, with boundaries taken from the original.
Markov	Chooses values from the list with a second order Markov chain.

Table 2. Functions that can be applied to pitch manipulation of recorded phrases.

Same	Bypasses the original list with no modification.
Ascending	Rearranges the list of PC in ascending order of values.
Descending	Rearranges the list of PC in descending order of values.
Transpose	Transpose pitches in the original list by a random number of semitones.
Multiply	Multiplies the PC values by a random factor.
Complement	Replaces the original PC set with its complement ⁶ .
Ascending Complement	Rearranges the list of the complementary set of PC in ascending order of values.
Descending Complement	Rearranges the list of the complementary set of PC in descending order of values.
Transpose Complement	Transpose PC in the complementary set by a random number of semitones.
Multiply Complement	Multiplies the PC values of the complementary set by a random factor.
Markov	Chooses values from the PC set with a second order Markov chain.

Table 3. Functions that can be applied to amplitude manipulation of recorded phrases.

Same	Bypasses the original list with no modification.
Crescendo	Rearranges the list of intensities in ascending order of values.
Decrescendo	Rearranges the list of intensities in descending order of values.
Equal	Chooses a duration and repeats it for the number of items in the list.

⁶ As defined by Forte^[16], when receiving a pitch class set equivalent to the C major scale (the white notes of the piano), for example, its complementary set would be the F sharp major pentatonic scale (the black notes of the piano), which represents all the pitch classes that remain for the universal set of twelve notes to be reached.

After each list of parameters is transformed by these functions, the contrapuntal method combines the parameters to form a sound event in a new list, that also contains a legato index⁷, directly replicated from the last phrase played by the musician.

A Frequency Modulation (FM) synthesizer receives the list containing the parameters set for the next sound event (timbre, pitch, intensity, duration, and legato index), combining these defined aspects with some degree of randomness in the synthesis process to permit subtle variations between each sound played in a phrase. The output of the FM synthesizer can be passed through a randomly selected sound effect, further modifying the resulting sound. The effects are a ring modulator, a pitch shifter, some delay lines, a granular synthesizer, low-pass and high-pass filters, an amplitude modulator, and a reverb. The parameter settings for each sound effect use the same list of parameters received by the synthesizer and randomized values selected for each event.

Network improvisation sessions can also be held using [REDACTED]. At the moment, this functionality makes it possible to interact aurally with a second user and their version of the system, geographically distant, in real-time, with very low latency by using *Quacktrip*⁸, a Pure Data implementation of the *JackTrip* audio transmission protocol. Although ubimus initiatives are not necessarily dependent on proposals based on synchronous musical interaction^[10], we note that Rottondi *et al.*^[16] suggest that situations of instrumental interaction can tolerate delays of between twenty and fifty milliseconds. According to Bosi *et al.*^[17], *JackTrip* allows latencies below two hundred milliseconds for connections made between points more than ten thousand kilometers apart, allowing for viable synchronicity between two parties if this is in their interests.

3. Ubimus and improvising machines

In this section, we explore the similarities between musical perspectives in the ubimus movement and those found in the principles behind the creation of musical agents, especially those designed for free improvisation. Examples of characteristics found in [REDACTED] and other improvising machines will illustrate these relationships.

⁷ Calculated from the proportion of silence between two notes, not identified as a pause.

⁸ To refer to the documentation, see <http://msp.ucsd.edu/tools/quacktrip/doc.htm>.

First, the ubimus movement aims to integrate musical and sonic materials, challenging the traditional separation found in many classical and electronic music practices^[2]. Similarly, many improvising machines, especially those developed for the context of free improvisation, seek to bring together timbral and symbolic elements as aspects of equal relevance to sound creation and interaction with the system. For example, systems developed by Hsu^[20] benefit from an analysis of spectral roughness measured from sound input alongside intensity and tempo to identify different “modes” of playing. Another relevant approach by Collins^[15] employs k-means clustering to identify ten different sonic scenarios based on the timbre content of the last few seconds. Nort et al.^[21] favored recognizing gestural and textural content in their free improvisation machine, emphasizing sonic elements as the musical basis for the interaction.

Furthermore, Lewis^[22] highlights that improvising machines should be viewed as active participants in music-making, rather than passive tools controlled by composers. This perspective aligns with ubimus principles, emphasizing non-hierarchical collaboration. A similar point of view is supported by the ubimus movement when authors indicate that “a paradigm of musical interaction targeting passive ‘instruments’, ‘orchestras’ or enforcing centralized decision-making may ignore key aspects of this ubimus ecosystem”^[23] and that in “the context of ubiquitous systems, a device is not a passive object that a musician can play. It is an agent in a dynamical system that adapts itself to the musical activity, to the local environment, and to the other agents that interact with it”^[1]. Illustrating this idea, the orchestra developed by Lewis in his early renditions of *Voyager* allowed for the independence of decisions made by each of the 64 voices available in the system^[24], eliminating the centralizing role of a conductor in favor of a model of multifaceted collective musical production, such as that found in the encounters of the AACM collective of which he was a fundamental participant^[22].

Many improvising machines, as well as [REDACTED], seek to analyze and manipulate symbolic elements of the material played by a musician with whom it interacts to produce their behavior. This paradigm, for Lewis, represents an interaction technique that is “essentially Eurocentric in nature”, as reintroductions of material that has already been played “come into conflict” with the direction of improvisation^[25]. Contrasting to this method, old renditions of *Voyager* employed the analysis of the material played by the musician only as a “filter” for the system’s decisions,

restricting itself to listening to a short-term memory consisting of only the last five notes played by the musician.

This “issue” is addressed in [REDACTED] by disassociating each parameter from the others found in a phrase, and by also transforming the material through mathematical operations to the numerical representation of each musical element of the phrases manipulated, as seen in the previous section. The decision-making process is also nondeterministic, applying second order Markov chains to navigate its memory of recorded phrases. The reappearance of any musical material, therefore, cannot be controlled by those who play with the system, who at most act as an influence on the results. The output produced by [REDACTED] also privileges sonic instead of symbolic responses (such as MIDI), synthesizing sounds and modifying them with digital effects of which some parameters are controlled by stochastic methods, further removing the musician’s control over the system’s behavior.

Keller *et. al*^[26] identify *ubimus* practices under the methodological framework of sonic ecologies: “habitats where agents and objects interact producing creative sonic by-products”. While Keller and Lazzarini^[27] stress that *ubimus* depend not only on musical and sound resources but on the relationships established between peers and with the surrounding environment. Free improvisation, while a practice particularly focused on the act of collective performance, embraces adaptive behaviors of participants to the audience and particular sonic characteristics of a venue. Improvising machines developed for this context can be thought of as agents capable of establishing creative relationships with the sonic environment that differ from those assembled by human performers.

Ubimus also embraces approaches to music-making that do not depend on acoustic instrumentation models, traditionally associated with corporeal movement and sometimes seen as the “only valid strategies for musical interaction”^[1]. This paradigm, in Keller’s^[28] view, ends up imposing interaction modes and processes developed and maintained by the European classical music tradition. Even though a large population within the genre of free improvisation still relies on the use of traditional acoustic instruments^[9], their interaction approaches and instrumental techniques often question those traditionally associated with classical music, allowing not only the presence of unusual techniques and sound sources but permitting the interaction of music with other forms of artistic expression. For example, Keith Rowe, the

guitarist for the group AMM, reports abandoning his guitar technique by ceasing to tune it and preferring to manipulate it with objects of various kinds with the instrument laid down on a table. The process was inspired by how Jackson Pollock abandoned traditional painting techniques in favor of the “drip technique”, as Rowe himself comes from a visual arts background^[29].

When dealing with technological interfaces, Keller^[28] also notes that the design of ubimus tools should provide intuitive interfaces, avoiding “cognitive overload” for users who are already busy maintaining their creative proposals. Observing this perspective, the development of [REDACTED] endeavored to make it easier for users to adjust basic settings needed to start a performance: its interface makes it possible to prepare and begin an improvisation session with just three mouse interactions. Similarly, an online session can be initiated with just two more mouse clicks.

Finally, as discussed by Scott Thomson^[30], free improvisation questions the role of the improviser as a virtuoso, contradicting a historical perspective of European concert music very recurrent in jazz cultures. In the same way, ubimus is based on creative approaches that embrace “little-c musical phenomena that were excluded from professional musical practices”^[26], questioning the preeminence of virtuosity in music-making.

4. Discussion and future perspectives

This section addresses prospects for [REDACTED], based on the theoretical relationships between improvising machines and the ubimus movement, presented in the previous section.

Originally designed as a companion for practicing free improvisation, [REDACTED] now demonstrates potential for diverse creative applications. We address the possible inclusion of audience participation and/or environmental resources, where distinct sonic and relational ecosystems from those of free improvisation might be achieved. Future developments could incorporate sonic relationships with the environment into the system's audio analysis. This approach, overlooked in the current design, aligns with ubimus perspectives by challenging the binary division between 'desired' and 'undesired' sounds in music^[31]. Another perspective could be brought by permitting that, if the system is used in a “concert” scenario (which it was not originally designed to), other modalities of interaction could be implemented by Local Area Network (LAN) connectivity.

Lewis^[32] points out that the distinction between the activity of “composition” and “improvisation”, “from the point of view” of a computer, does not exist. So how do we perceive the difference between the two perspectives? For him, the notion of algorithmic composition stems from the idea that the output of a composition program does not allow for any result, but only the “correct” ones. In this sense, the use of [REDACTED] for other initiatives focused on a compositional approach could also be explored. Since a recording feature is already available for both sounds played by the musician, as well as the sounds produced by the machine, each improvisation recorded with the system generates sound material to be used in electroacoustic composition proposals, or sonic ideas to be explored in compositions or comprovisations of instrumental or interactive pieces.

Although the preferred model for interaction in free improvisation is usually by directly approaching sound relations, possibilities can also be explored through the MIDI output provided by [REDACTED]. For example, instruments that allow both the reproduction of sounds through their own speakers and the exchange of MIDI data with a computer, such as most commercially available digital keyboards, could be explored in pieces whose system response uses the instrument played by the musician. Although this is a process of interaction, modifications to [REDACTED]'s response algorithms could facilitate the perception that this is an “extended instrument” with its own personality, the control of which is not entirely in the hands of the player.

The same proposals can be imagined for other FLOSS musical agents that permit their users the art of modifying their code, allowing for creative outcomes unforeseen by their original developers. The DIY nature of our project can be viewed as both an inspiration and an invitation to interested stakeholders to take part in the development of novel creative applications. As the presence of commercial solutions that employ the use of artificial intelligence (AI) becomes increasingly ubiquitous, there is a need to reflect on the impact of the usual nature of these ventures, which are usually closed-source and centralized in for-profit companies. We have observed that, in general, their users have little understanding of how they work, the limits of these intelligences, and the treatment and purpose of the data generated by interaction with these systems. Creative proposals of a FLOSS nature offer a different way of addressing the interaction

with intelligent systems, functioning not only as a "one size fits all" tool but also allowing its adaptation to different social contexts and their specific local resources.

5. Conclusion

This paper explored the intersection of the ubimus movement and improvising machines, presenting [REDACTED], an interactive system designed for free improvisation. By examining the theoretical relationships between these domains, we illustrated how both seek to break down traditional boundaries in music-making, embracing non-hierarchical, adaptive, and collaborative approaches to creativity.

Both ubimus and free improvisation rely on dynamic relationships between agents, peers, and the surrounding environment, creating ecosystems where interaction and adaptation are key. Hence, we envision that improvising machines could have the potential for various artistic endeavors in both scenarios. [REDACTED], designed for accessibility and adaptability, already facilitates innovative approaches to improvisation and other creative musical activities involving artificial agents. By taking advantage of network connectivity, we have a broader scope of possible interactions between this project and other ecosystems in the *musical internet*.

In conclusion, [REDACTED] offers a model for creating adaptive, non-hierarchical, and community-oriented musical environments. Its FLOSS nature enhances its potential for widespread use and continual improvement, encouraging users to innovate and adapt the system according to their unique desires and experiences, in contrast with the closed and often opaque commercial AI solutions that have been dominating the creative technology landscape.

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